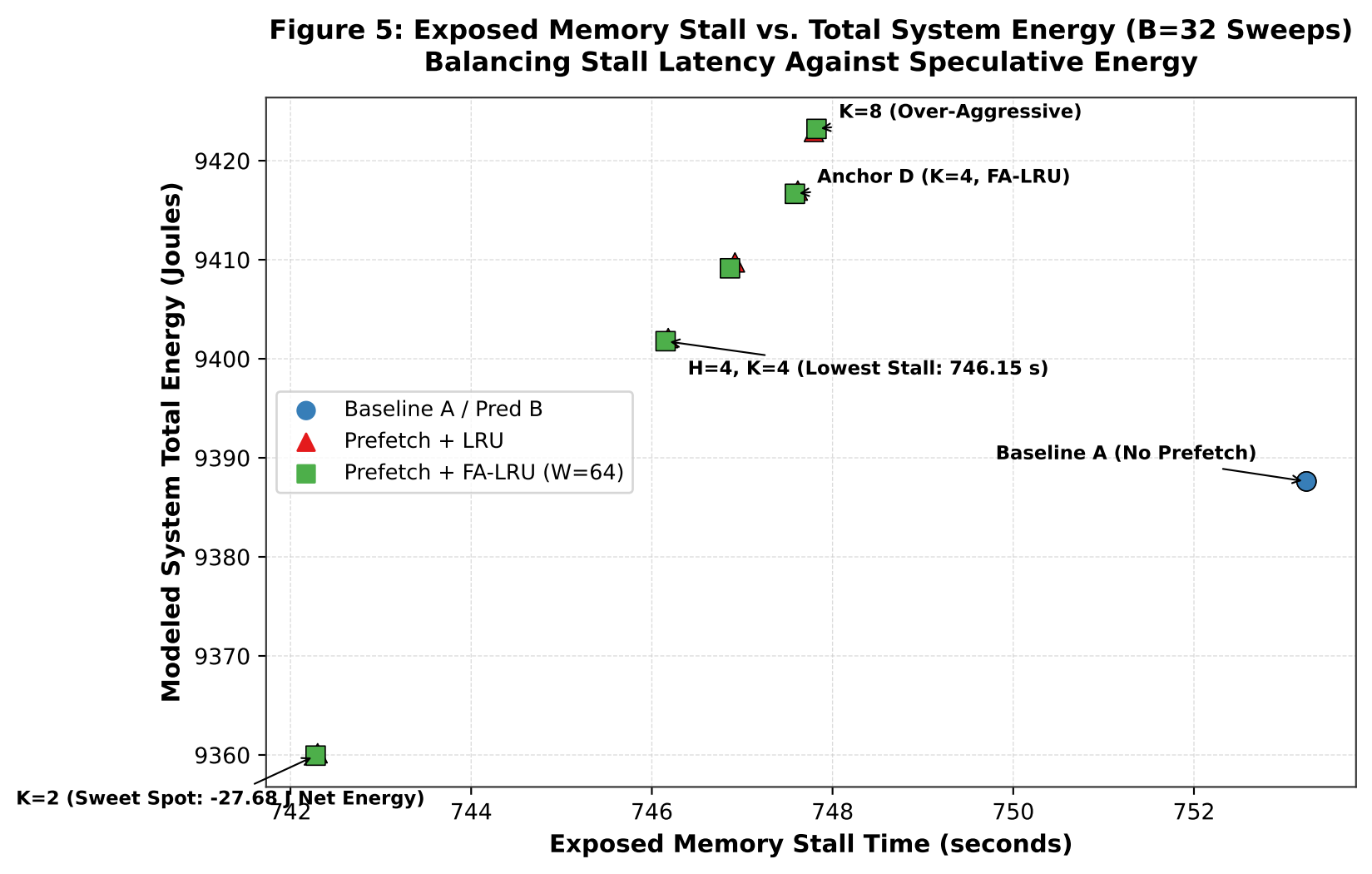
Figure 5: Exposed Memory Stall vs. Total System Energy (B=32 Sweeps)
Balancing Stall Latency Against Speculative Energy
K=8 (Over-Aggressive)
Anchor D (K=4, FA-LRU)
H=4, K=4 (Lowest Stall: 746.15 s)
Baseline A (No Prefetch)
K=2 (Sweet Spot: -27.68 J Net Energy)
Baseline A / Pred B
Prefetch + LRU
Prefetch + FA-LRU (W=64)
Modeled System Total Energy (Joules)
9420
9410
9400
9390
9380
9370
9360
742
744
746
748
750
752
Exposed Memory Stall Time (seconds)